

Math 3IA3 - Assignment 3

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Exercise 3.4.6

Let $x_n := n^{\frac{1}{n}}$ for $n \in \mathbb{N}$.

- (a) Show that $x_{n+1} < x_n$ if and only if $(1 + \frac{1}{n})^n < n$, and infer that the inequality is valid for $n \geq 3$. Conclude that (x_n) is ultimately decreasing and that $x := \lim_{n \rightarrow \infty} (x_n)$ exists.

Solution:

- (I) Show that $x_{n+1} < x_n \iff (1 + \frac{1}{n})^n < n$.

Using basic algebraic manipulations, we find:

$$\begin{aligned}
 x_{n+1} &< x_n \\
 &\iff \\
 (n+1)^{\frac{1}{n+1}} &< n^{\frac{1}{n}} \\
 &\iff \\
 ((n+1)^{\frac{1}{n+1}})^{n+1} &< (n^{\frac{1}{n}})^{n+1} \\
 &\iff \\
 n+1 &< (n^{\frac{1}{n}})^n \cdot n^{\frac{1}{n}} \\
 &\iff \\
 n+1 &< n \cdot n^{\frac{1}{n}} \\
 &\iff \\
 \frac{n+1}{n} &< n^{\frac{1}{n}} \\
 &\iff \\
 1 + \frac{1}{n} &< n^{\frac{1}{n}} \\
 &\iff \\
 \left(1 + \frac{1}{n}\right)^n &< (n^{\frac{1}{n}})^n \\
 &\iff \\
 \left(1 + \frac{1}{n}\right)^n &< n
 \end{aligned}$$

Therefore, $x_{n+1} < x_n \iff (1 + \frac{1}{n})^n < n$

- (II) Show that $(1 + \frac{1}{n})^n < n$, $\forall n \geq 3$.

We will proceed to prove $(1 + \frac{1}{n})^n < n$, $\forall n \geq 3$ via induction. Let $P(n)$ be the statement $P(n) : (1 + \frac{1}{n})^n < n$.

Base Case:

LHS:	RHS
$= (1 + \frac{1}{n})^n$	$= n$
$= \left(1 + \frac{1}{3}\right)^3$	$= 3$
$= \left(\frac{4}{3}\right)^3$	$= 3$
$= \frac{64}{27}$	$= 3$

Therefore, since $\text{LHS} < \text{RHS}$, the base case holds and $P(3)$ is true.

State Induction Hypothesis:

Let $k \in \mathbb{N}$ be arbitrary, we can therefore formulate our induction hypothesis as:

$$P(k) : (1 + \frac{1}{k})^k < k$$

Prove $P(k) \implies P(k+1)$:

$$\begin{aligned}
 (1 + \frac{1}{k+1})^{k+1} &= (1 + \frac{1}{k+1})^k \cdot (1 + \frac{1}{k+1}) \\
 &\leq (1 + \frac{1}{k})^k \cdot (1 + \frac{1}{k}) \quad \langle \text{Since } \frac{1}{k+1} < \frac{1}{k} \forall k \in \mathbb{N} \rangle \\
 &< k \cdot (1 + \frac{1}{1+k}) \quad \langle \text{By induction hypothesis} \rangle \\
 &= k+1
 \end{aligned}$$

Therefore, $P(k) \implies P(k+1)$.

Since $P(k) \implies P(k+1)$ for arbitrary $k \in \mathbb{N}$, and since we know $P(3)$ is true, then by principle of mathematical induction, we know that $P(n)$ holds $\forall n \in \mathbb{N}$ such that $n \geq 3$.

(III) Show $x := \lim_{n \rightarrow \infty} (x_n)$ exists.

From parts (I) and (II), since $x_{n+1} < x_n \iff (1 + \frac{1}{n})^n < n$, and $(1 + \frac{1}{n})^n < n$ holds for all $n \geq 3$, we have that $x_{n+1} < x_n$ is true $\forall n \geq 3$ and thus the sequence x_n is eventually monotone. Furthermore, since $n \geq 1$, we know that $(n)^{\frac{1}{n}} \geq 1$ and thus, since (x_n) is monotonously

decreasing past $n = 3$ and is lowerbounded by 0, by the Monotone Convergence Theorem, we know that (x_n) converges.

(b) Use the fact that the subsequence (x_{2n}) also converges to x to conclude that $x = 1$.

Solution:

Since we know that (x_n) converges, we know that all subsequences of (x_n) also converge to x . Therefore, using the limit identity proven in the textbook that $\lim_{n \rightarrow \infty} (p)^{\frac{1}{n}} = 1 \ \forall p \in \mathbb{R}, r \geq 1$, we can find that:

$$\begin{aligned}
 \lim_{n \rightarrow \infty} \left((2n)^{\frac{1}{2n}} \right) &= x \implies \\
 \lim_{n \rightarrow \infty} \left((2n)^{\frac{1}{n}} \right) &= x^2 \implies \\
 \lim_{n \rightarrow \infty} \left(2^{\frac{1}{n}} n^{\frac{1}{n}} \right) &= x^2 \implies \\
 \lim_{n \rightarrow \infty} \left(2^{\frac{1}{n}} \right) \cdot \lim_{n \rightarrow \infty} \left(n^{\frac{1}{n}} \right) &= x^2 \implies \\
 1 \cdot x &= x^2 \implies \\
 x &= x^2 \implies \\
 x^2 - x &= 0 \implies \\
 x(x - 1) &= 0
 \end{aligned}$$

Therefore, $x = 1$ or $x = 0$, but since we know that $x_n \geq 1 \ \forall n \in \mathbb{N}$, x must equal 1. Therefore, $\lim_{n \rightarrow \infty} (x_n) = 1$.

Exercise 3.5.5

If $x_n := \sqrt{n}$, show that (x_n) satisfies $\lim_{n \rightarrow \infty} |x_{n+1} - x_n| = 0$ but is not a Cauchy sequence.

Solution:

(I) Show that x_n satisfies $\lim_{n \rightarrow \infty} |x_{n+1} - x_n| = 0$.

Taking it as given that the sequence $\frac{1}{n}$ converges to zero, by algebraic limit rules, we can determine that the sequence $\frac{1}{\sqrt{n}}$ also converges to zero. Thus, for arbitrary $\epsilon > 0$, we know that there exists some $N \in \mathbb{N}$ such that $\forall \geq N, \frac{1}{\sqrt{n}} < \epsilon$. Therefore, for all such n , we have that:

$$\begin{aligned}
 |(x_{n+1} - x_n) - 0| &= |\sqrt{n+1} - \sqrt{n}| \\
 &= \sqrt{n+1} - \sqrt{n} \quad \langle \text{Since } \sqrt{n+1} > \sqrt{n} \forall n \in \mathbb{N} \rangle \\
 &= \sqrt{n+1} - \sqrt{n} \cdot \frac{\sqrt{n+1} + \sqrt{n}}{\sqrt{n+1} + \sqrt{n}} \\
 &= \frac{(\sqrt{n+1} - \sqrt{n})(\sqrt{n+1} + \sqrt{n})}{\sqrt{n+1} + \sqrt{n}} \\
 &= \frac{n+1 - n}{\sqrt{n+1} + \sqrt{n}} \\
 &= \frac{1}{\sqrt{n+1} + \sqrt{n}} \\
 &\leq \frac{1}{\sqrt{n}} \\
 &< \epsilon
 \end{aligned}$$

Thus, since for arbitrary $\epsilon > 0$, we can find an $N \in \mathbb{N}$ such that for all $n \geq N, |x_{n+1} - x_n| < \epsilon$, by definition of convergence, $\lim_{n \rightarrow \infty} |x_{n+1} - x_n| = 0$.

(II) Show that x_n is not a Cauchy sequence.

We show that x_n is not a Cauchy sequence by proving that it diverges because, over the set of real numbers, a sequence is Cauchy if and only if it is convergent to a real number. Let $M^2 > 0$ be arbitrary. By the Archimedean property, we know that there exists an $N \in \mathbb{N}$ such that for all $n \geq N \implies n > M^2$. Thus:

$$\begin{aligned}n &\geq N \implies \\n &> M^2 \implies \\\sqrt{n} &> \sqrt{M^2} \implies \\x_n &> M\end{aligned}$$

Therefore, we know that $\lim_{n \rightarrow \infty} (x_n)$ diverges to infinity and therefore (x_n) cannot be Cauchy.

Exercise 3.6.10

Show that if $\lim_{n \rightarrow \infty} \left(\frac{a_n}{n}\right) = L$, where $L > 0$, then $\lim_{n \rightarrow \infty} (a_n) = +\infty$.

Solution:

Let $M > 0$ be arbitrary. Since $\lim_{n \rightarrow \infty} \frac{1}{n} = 0$ is known, we can conclude using algebraic limit rules that $\lim_{n \rightarrow \infty} \left(\frac{M}{n}\right) = 0$. Since $\lim_{n \rightarrow \infty} \left(\frac{a_n}{n}\right) = L > 0$, we know there exists some $N_0 \in \mathbb{N}$ such that $\forall n \geq N_0$:

$$\left|\frac{a_n}{n} - L\right| < \frac{L}{2}$$

And therefore, $\frac{L}{2} < \frac{a_n}{n} < \frac{3}{2}L$ for all such n . Likewise, we know that there exists some $N_1 \in \mathbb{N}$ such that $\forall n \geq N_1$:

$$\left|\frac{M}{n}\right| < \frac{L}{2}$$

And therefore, $0 < \frac{M}{n} < \frac{L}{2}$ for all such n . Thus, letting $N = \max\{N_0, N_1\}$, we have that $\forall n \geq N$:

$$\begin{aligned} \frac{a_n}{n} &> \frac{M}{n} \\ a_n &> M \end{aligned}$$

Therefore, since for arbitrary $M > 0$, we can always find an $N \in \mathbb{N}$ such that $a_n > M \forall n \geq N$, $\lim_{n \rightarrow \infty} (a_n) = +\infty$.

Exercise 3.7.11

If $\sum_{n=1}^{\infty} a_n$ with $a_n > 0$ is convergent, then is $\sum_{n=1}^{\infty} (a_n)^2$ always convergent? Either prove it or give a counter-example.

Solution:

Proof. We know, by definition of infinite sums, that if $\sum_{n=1}^{\infty} a_n$ convergent to some limit a , then the sequence of its partial sums, defined by $(s_k) = \sum_{n=1}^k (a_n)$ must converge likewise converge to a . Therefore, we know that $\lim_{k \rightarrow \infty} (s_k) = a$. Using the same logic, we know that the infinite sum $\sum_{n=1}^{\infty} (a_n)^2$ converges if the sequence of its partial sums converges. Since $a_n > 0$, $\forall n \in \mathbb{N}$, we know that (s_k) is increasing and since $a_n > 0 \implies (a_n)^2 > 0$, we likewise know the sequence of partial sums of $(a_n)^2$ is likewise increasing. Furthermore, as a corollary to the fact (s_k) is increasing, we know that a must upperbound (s_k) . Now, algebraic limit rules dictate that $(\lim_{k \rightarrow \infty} (s_k))^2 = a^2$, however, notice that:

$$\begin{aligned}
 (s_k^2) &= \left(\sum_{n=1}^k (a_n) \right)^2 \\
 &= \sum_{n=1}^k (a_n) \cdot \sum_{n=1}^k (a_n) \\
 &= (a_1 + a_2 + \dots + a_k) \cdot (a_1 + a_2 + \dots + a_k) \\
 &= a_1^2 + a_1 a_2 + \dots + a_1 a_k + a_2 a_1 + a_2^2 + \dots + a_2 a_k + \dots + a_k a_1 + a_k a_2 + \dots + a_k^2 \\
 &\geq a_1^2 + a_2^2 + \dots + a_k^2 \quad \langle \text{Since } a_n > 0 \forall n \in \mathbb{N} \rangle \\
 &= \sum_{n=1}^k (a_n)^2
 \end{aligned}$$

Therefore, we know that the sequence of partial sums of $0 < \sum_{n=1}^k (a_n)^2 \leq s_k$, we therefore know that a must also upperbound the sequence of partial sums of $(a_k)^2$. Thus, since the sequence of partial sums of $(a_k)^2$ is both increasing and upperbounded by a , by Monotone Convergence Theorem, the sequence converges and thus, the infinite sum $\sum_{n=1}^{\infty} (a_n)^2$ converges. $\square$

Exercise 4.1.7

Show that $\lim_{x \rightarrow c} (x^3) = c^3$ for any $c \in \mathbb{R}$.

Solution:

We can easily show this using the sequential definition of functional limits. Let (x_n) be any sequence defined on $\mathbb{R}$ (since the domain of x^3 is $\mathbb{R}$) such that $x_n \neq c \ \forall n \in \mathbb{N}$ and $\lim_{n \rightarrow \infty} (x_n) = c$. By the algebraic limit theorems, we know that:

$$\begin{aligned} \lim_{n \rightarrow \infty} (x^3) &= \left(\lim_{n \rightarrow \infty} (x_n) \right)^3 \\ &= c^3 \end{aligned}$$

Therefore, by definition of functional limits, since for all such (x_n) , $\lim_{n \rightarrow \infty} (x_n)^3 = c^3$, $\lim_{x \rightarrow c} (x)^3 = c^3$.